

Signals & Systems (EE2008)

Date: April 6 2024

Course Instructor

Mr. Dr. S.M. Sajid

Mr. Asim Hussain Farooqi

Sessional-2

Total Time: 1 Hours

Total Marks: 30

Total Questions: 03

Semester: SP-2024

Campus: Lahore

Haseeb ur Rehman
Student Name

22L-6234
Roll No

4A
Section

Haseeb
Signature

Q. No 1: Analyze the square pulse periodic signal given below in Fig-1, calculate all the coefficients of trigonometric fourier series and also write the final expression of trigonometric fourier series.

[CLO No. 3] [10]

$T_0 = 2\pi$

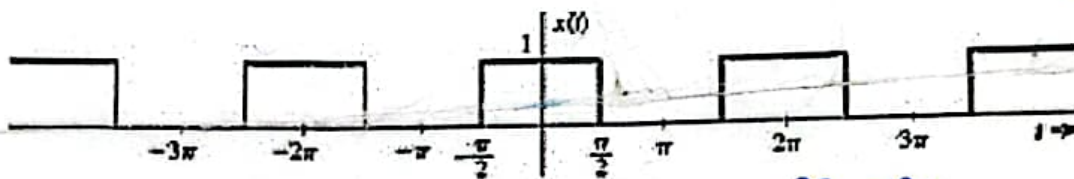


Fig-1

$$\frac{2\pi}{2\pi} = 0$$

$$\frac{2\pi}{\pi/2}$$

$$\frac{1}{2\pi} \int_{-\pi}^{\pi} x(t) dt$$

$$\frac{1}{2\pi} \times 2\pi$$

$$\frac{1}{2}$$

$$\frac{2\pi}{2}$$

Q. No 2 Analyze the periodic signal shown in Fig-2 with period $T_0 = 2$, find the exponential fourier series coefficient and sketch the exponential fourier spectra.

[CLO No. 3] [15]

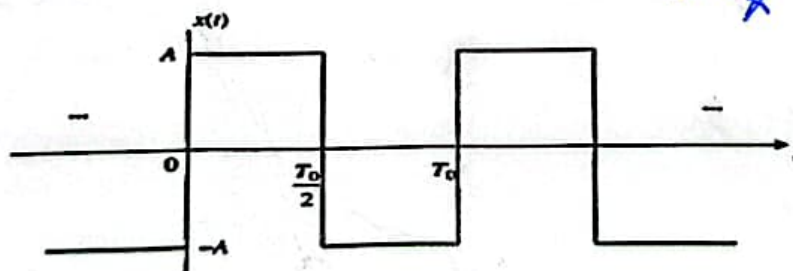


Fig-2

$$2\pi \times \frac{2}{\pi}$$

$$\frac{\pi}{\pi} = 0$$

Q. No 3: Find the fourier transform for $g(t) = \text{rect}\left(\frac{t}{\tau}\right)$ and analyze by sketching the $G(\omega)$.

[CLO No.4] [5]

$$t/\tau \rightarrow G(\omega)$$

National University of Computer and Emerging Sciences
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Signals & Systems (EE2008)

Final Exam

Date: May 22, 2024

Course Instructor(s)

Mr. Dr. S.M. Sajid

Mr. Asim Hussain Farooqi

Total Time (Hrs):

3

Total Marks:

100

Total Questions:

6

22L-6234

Roll No

4A

Section

Haseeb Hossain
Student Signature

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Attempt all the questions.

CLO # 1: Express different types of signals and systems.

Q1: A continuous-time signal $x(t)$ is shown in Fig-1 below. Compute and label carefully each of the following signals: [20]

a) $x(2t + 1)$

b) $[x(t) + x(-t)]u(t)$

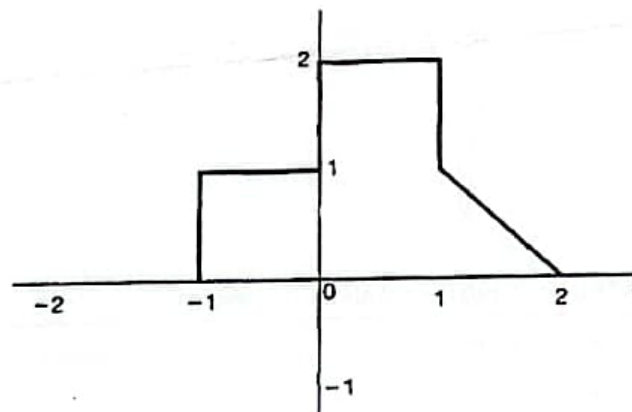
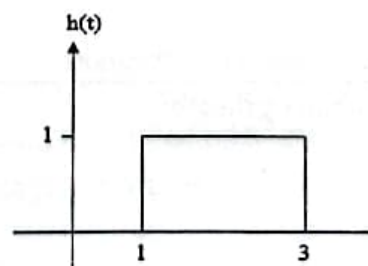
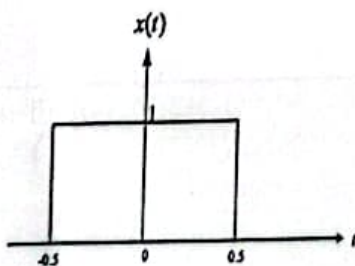


Fig-1

CLO # 2: Apply mathematical and graphical convolution techniques to analyze CT and DT systems.

Q2: Solve the convolution by graphical method, the signal $x(t)$ and $h(t)$ are given below. [20]



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CLO # 3: Analyze signals using Fourier series representation.

Q3: Analyze the periodic signal shown in Fig-2, calculate all the coefficients of trigonometric Fourier series and also write the final expression of trigonometric Fourier series. [20]

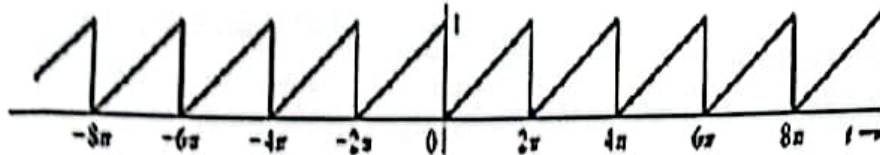
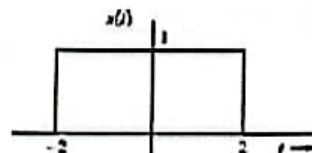


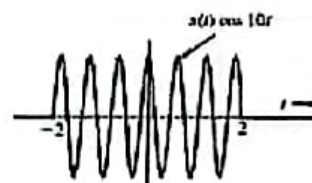
Fig-2

CLO # 4: Analyze signals using Fourier transform and its properties.

Q4: Analyze the signal given below, find and sketch the Fourier transform of the modulated signal $x(t)\cos(10t)$ in which $x(t)$ is a gate pulse $\text{rect}\left(\frac{t}{4}\right)$ as illustrated in figure given below. [15]

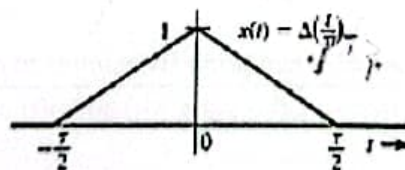


(a)



CLO # 4: Analyze signals using Fourier transform and its properties.

Q5: Analyze the signal given below, find and sketch the Fourier transform of $\Delta\left(\frac{t}{T}\right)$ clearly mention the method used or the property applied. [15]



CLO # 5: Describe and use the sampling theorem.

Q6: State and prove the sampling theorem.

[10]